

Hor Wen Xuan (Alex)

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EDUCATION

SINGAPORE MANAGEMENT UNIVERSITY

Aug 2025 - Mar 2027

Master of IT in Business (Artificial Intelligence Track) – CGPA: 3.83/4.00

- Modules Taken: Applied Machine Learning, Generative AI with Large Language Models, Modern Software Solutions, Machine Learning Engineering

NANYANG TECHNOLOGICAL UNIVERSITY

Aug 2017 - May 2021

Bachelor of Communication Studies (Communications Research) - Honours with Distinction

EXPERIENCE

HOME TEAM SCIENCE AND TECHNOLOGY AGENCY (HTX) – Enterprise AI Products Intern

May 2026 - Current

- Conducting technical discovery and capability evaluation for an enterprise Speech-to-Text and Generative AI product for Ministry of Home Affairs (MHA) agencies.

TRI-SECTOR ASSOCIATES - Associate, Singapore

Jul 2023 - Nov 2024

- Built a market sizing model by synthesizing educational and salary data for a financial startup, resulting in an award of 1M USD from a global bank.
- Secured \$500K in funding for a government-supported program by pricing an innovative outcomes-based guarantee based on data analysis of program track record.

MCKINSEY & COMPANY - Project Analyst, Singapore (Contract Role)

Jun 2022 - Feb 2023

- Synthesized cost-saving analytics and cannibalization risks into a \$20M pricing model to secure stakeholder alignment for a digital product launch.

ACCENTURE - Technology Analyst, Singapore

Jul 2021 - Jun 2022

- Designed a national billing system by centralizing business processes across 3 healthcare clusters, reducing maintenance costs by 25%.

TECHNICAL PROJECTS

Modern Software Solutions: Credit Card Optimiser

- Developed a RESTful API using FastAPI to manage user wallet states, implementing CRUD logic with modular Python architecture.
- Integrated OpenAI's API for recommendation synthesis; implemented automated unit testing (Pytest) and Git-based version control to maintain CI/CD readiness.

Generative AI with LLMs: Evaluating Agentic Frameworks for Financial Q&A Over SEC Filings

- Evaluated the performance of agentic frameworks (LangGraph, CrewAI, LlamaIndex) for financial question-answering on SEC filings, identifying that LangGraph achieved the highest correctness (0.665).

Machine Learning Engineering: Automated Credit-Default Prediction Pipeline

- Built an end-to-end PySpark/Airflow ML pipeline (Medallion architecture) training a LightGBM credit-default model to 0.817 out-of-time AUC (63% lift), with a versioned MLflow model registry.

ADDITIONAL

- Programming Languages:** Python, SQL, PySpark, Pandas, NumPy.
- AI & Machine Learning:** Agentic Workflows (LangGraph, CrewAI, LlamaIndex), Generative AI Engineering (LLM-as-a-Judge, Fine-tuning), Applied ML (SHAP Interpretability).
- Software Engineering & Infrastructure:** REST API Development (FastAPI), CI/CD Readiness (Git, Pytest, Agile), Pipeline Orchestration (Airflow).